

Project Data Dictionary and Metric Definitions

This document describes the workbook created by the notebook and the safest interpretation of its measures. The row-level `Scored_Customers` sheet is the primary customer table. `Model_Comparison` and `Threshold_Analysis` contain test-set evaluation summaries and should remain separate from the full-customer scoring table.

Workbook sheets

Sheet	Grain	Use
<code>Scored_Customers</code>	One row per customer	Customer filters, descriptive churn analysis, risk segmentation, and score review
<code>KPI_Summary</code>	One row for the full file	Reconciliation reference; recreate measures from <code>Scored_Customers</code> for filter-responsive cards
<code>Segment_Summary</code>	One row per Contract × InternetService combination	Reconciliation/reference summary; prefer dynamic measures from <code>Scored_Customers</code> for slicer-responsive visuals
<code>Model_Comparison</code>	One row per candidate model	Compare held-out test-set metrics
<code>Threshold_Analysis</code>	One row per probability threshold	Show held-out test-set precision, recall, F1, and workload trade-offs
<code>Risk_Summary</code>	One row per operational risk tier	Reconcile tier counts, observed churn, scores, and monthly charges
<code>Reconciliation_Log</code>	One row per quality check	Verify record counts, null checks, and key totals

`Scored_Customers` fields

Field	Meaning	Use / caution
<code>customerID</code>	Customer identifier	Count distinct IDs; do not use as a predictor or chart category

Field	Meaning	Use / caution
gender , SeniorCitizen , Partner , Dependents	Demographic/account profile fields	Use only for descriptive breakdowns relevant to the project; avoid implying causation
tenure	Customer tenure in months	Numeric value; useful for lifecycle analysis
TenureGroup	Bands: 0–6, 7–12, 13–24, 25–48, 49–72 months	Use for readable churn-rate charts; keep categories in this natural order
Contract	Month-to-month, one-year, or two-year contract	Key descriptive segment
InternetService	DSL, fiber optic, or no internet service	Key descriptive segment
PaymentMethod	Customer payment method	Electronic check has the highest observed churn rate in this dataset
PaperlessBilling	Whether paperless billing is enabled	Descriptive attribute only
NumAddonServices	Count of selected add-on services marked Yes	Derived feature; includes OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies
MonthlyCharges	Monthly charge in the source data	A recurring-charge measure; total by high-risk group is a monthly-charge proxy, not expected loss
TotalCharges	Accumulated charges, converted to numeric	Highly related to tenure; do not interpret its model effect independently without care
AvgMonthlySpend	$\text{TotalCharges} / \text{tenure}$, with current monthly charge used at zero tenure	Derived field close to MonthlyCharges ; ablation results show only small changes on this split
Churn	Historical outcome, Yes or No	Actual observed outcome, not a prediction
ChurnFlag	1 if Churn is Yes, else 0	Binary version of historical outcome; mean equals actual churn rate
ChurnProbability	Selected model's estimated score for a customer	Continuous score; calibration needs separate assessment, especially after SMOTE

Field	Meaning	Use / caution
PredictedChurnFl ag50	1 when ChurnProbability >= 0.50	Model classification at 50%; threshold- dependent
PredictedChurnFl ag60	1 when ChurnProbability >= 0.60	Threshold flag; aligns with High tier when tier boundaries are applied as documented
RiskTier	Low below 30%; Medium 30% to under 60%; High 60% or above	Operational reporting bands, not validated action rules

Core measures

Measure	Definition	Meaning
Actual churn rate	Churned customers ÷ customers in current filter context	Historical share that churned
Average predicted probability	Average of ChurnProbability in current filter context	Mean model score for these records; not the historical churn rate
Predicted churn rate @ 50%	Share with PredictedChurnFlag50 = 1	Model-flagged share at the 50% threshold
Predicted churn rate @ 60%	Share with PredictedChurnFlag60 = 1	Model-flagged share at the 60% threshold
High-risk rate	Share with RiskTier = High	High operational-tier share; should match the 60% threshold share if bins and flag are consistent
High-risk monthly charges	Sum of MonthlyCharges for High-tier customers	Monthly charges associated with high-risk records; not forecast revenue loss or expected customer lifetime value
Precision	True positive predictions ÷ all positive predictions	Share of flagged customers who actually churned, evaluated on the test set
Recall	True positive predictions ÷ all actual churners	Share of test-set churners detected
F1	Harmonic mean of precision and recall	Single summary of the precision/recall balance

Measure	Definition	Meaning
AUC	Area under ROC curve	Ranking discrimination across thresholds; not calibration
Brier score	Mean squared difference between predicted probabilities and outcomes	Lower is better for probability error, but alone does not prove calibration

Evaluation versus scoring

The model-comparison and threshold sheets are derived from the **1,409-row untouched test set**. The `Scored_Customers` sheet contains all **7,043** customers, including training records. Keep these grains visibly separate in Power BI: use the test-set sheets to discuss model performance, and the full-customer sheet for historical segmentation and scoring.

Notebook values to reconcile

- Customers: 7,043
- Historical churners: 1,869
- Actual churn rate: 26.5%
- Test set size: 1,409
- Selected model: Logistic Regression
- Selected-model test metrics: recall 0.794; precision 0.510; AUC 0.843; F1 0.621; Brier score 0.166
- High-risk tier: 2,298 customers; observed churn rate 57.57%; average predicted probability 77.95%